

MUSTAFA DOGAN

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EDUCATION

Hacettepe University 2021-2024
Master's in Science in Computer Engineering | GPA: 3.64/4.0
Ankara, Turkey
Graduate Thesis: [Evaluating Zero-Shot Learning Capabilities of Vision-Language Models](#)
Advisors: Erkut Erdem, Aykut Erdem

Konya Food and Agriculture University (KFAU) 2016-2021
Bachelor's in Science in Computer Engineering | GPA: 3.72/4.0
Konya, Turkey
Graduation Project: *Autonomously Navigated Agent Using Reinforcement Learning*
Advisor: Kasim Oztoprak

ACADEMIC RESEARCH EXPERIENCE

Fatima Fellowship Sep 2024-Jun 2025
Tarek Naous, Georgia Institute of Technology Remote
– To Lie or Not to Lie? Investigating The Biased Spread of Global Lies by LLMs: [ACL'26 Main](#)

Hacettepe University Computer Vision Lab Feb 2022-Jun 2024
Prof. Erkut Erdem, Prof. Aykut Erdem, Hacettepe University Ankara, Turkey
– FewMMBench: A Benchmark for Multimodal Few-Shot Learning: [arXiv'26](#) 🧡 [mustafaa/FewMMBench](#)
– Zero-Shot Linguistic Evaluation of Video Language Models: [ICLR'24](#)

Network and Architecture Research Lab Sep 2019- Jun 2021
Prof. Kasim Oztoprak, Konya Food and Agriculture University Konya, Turkey
– Teaching Computer Architecture from Bits and Bytes: [PeerJ Comp. Sci. Journal'24](#) (Peer Reviewed)
– Speculator and Influencer Analysis in Stock Market Using Social Media: [IEEE Big Data'20](#)

Food Sensory Lab Aug 2018
Prof. Sungeun Cho, Michigan State University MI, USA
– Project: Exploring Sleep-Eating Habits Using Python and Bayesian Analysis for Industry Insights

INDUSTRY RESEARCH EXPERIENCE

Aselsan Nov 2021 - Present
Mid-Level Software Engineer Ankara, Turkey
◦ Developed end-to-end firing control software for multiple air-defense platforms in modern C++, integrating distributed subsystems via Serial, Ethernet, CAN, and DDS.
◦ Developed on-premises LLM-based developer tools for code review, software assistance, and knowledge retrieval, leveraging RAG and agentic workflows.

Morten Inc. Jun - Sept 2020
Machine Learning Engineer Intern Istanbul, Turkey
◦ Created a *Knowledge Graph* to catalog authors and citations using *GPT-2*, *Python*, and *MongoDB*.
◦ Built a *GPT-2* to application to identify future tasks in academic papers, aiding in paper clustering.

PUBLICATIONS & PRE-PRINTS

ACL Main 2026 [To Lie or Not to Lie? Investigating The Biased Spread of Global Lies by LLMs](#)
Z Khan*, **M Dogan***, I Okoh, P Sadeghi, S Shrestha, SJ Nyah, MO Mokhiamar, MJ Ryan, T Naous

arXiv 2026 [FewMMBench: A Benchmark for Multimodal Few-Shot Learning](#)
M Dogan, I Kesen, I Calixto, A Erdem, E Erdem

ICLR 2024 [ViLMA: A Zero-Shot Benchmark for Linguistic and Temporal Grounding in Video-Language Models](#)
I Kesen, A Pedrotti, **M Dogan**, M Cafagna, EC Acikgoz, L Parcalabescu, I Calixto, A Frank, A Gatt, A Erdem, E Erdem

PeerJ Comp. Sci. [Teaching computer architecture by designing and simulating processors from their bits and bytes](#)
2024 **M Dogan**, K Oztoprak, MR Tolun

IEEE Big Data [Speculator and Influencer Evaluation in Stock Market by Using Social Media](#)
2020 **M Dogan**, O Metin, E Tek, S Yumusak, K Oztoprak

OPEN SOURCE SOFTWARE/DATA

GlobalLies  [zohaib-khan5040 / globallies](#)

GlobalLies is a multilingual, parallel dataset of misinformation prompts and entities designed to systematically evaluate how LLMs generate and propagate misinformation across different languages and countries.

FewMMBench  [mustafaa/FewMMBench](#)

FewMMBench is a benchmark designed to evaluate the few-shot learning capabilities of MLLMs under ICL and CoT prompting across a diverse set of multimodal understanding tasks involving image-text inputs.

ViLMA Benchmark  [ilkerkesen/ViLMA](#)

ViLMA benchmarks Video-Language Models in linguistic and temporal grounding across five dimensions: action counting, situation awareness, state change, rare actions, and spatial relations.

DoMe Architecture Simulator  [omer-metin/CPU16-Simulator](#)  [mustafaadogan/RISC16-Verilog](#)  [YouTube Playlist](#)

The DoMe architecture features a custom instruction set and data-path with 18 essential instructions for memory and register operations. A user-friendly desktop simulator, built in Python with PyQt, enables execution of assembly programs while displaying real-time memory and register values. The design and functionality were verified through a Verilog implementation in the ISE development environment.

Stock Market Analysis Datasets  [Tweet Dataset](#)  [NASDAQ Value Dataset](#)  [omer-metin/TweetCollector](#)

This project utilizes two datasets: one consisting of 3.4 million NASDAQ-related tweets, which helps identify market speculators and influencers by collecting tweet content, user information, and engagement metrics through Selenium (Tweet-Collector) and storing them in an SQLite database, and another containing daily market share value changes for Amazon, Apple, Google, Microsoft, and Tesla from mid-2010 to mid-2020.

VOLUNTEERING ACTIVITIES

Erasmus Youth Exchange Project, *Interstellar Inclusion*

Oct 2019

Participant

Perugia, Italy

- Actively engaged in fostering understanding and combat *racism* and *xenophobia* by peer experiences.
- Collaborated in a theater performance with a diverse group to address educational access challenges.
- Created a short movie on Freedom, winning the 'Best Movie' accolade among five nominees.  [Movie](#)
- Delivered a presentation on migration in Turkey to 30+ youngsters from diverse backgrounds.

Erasmus Youth Exchange Project, *Sail Entrepreneurship*

May 2018

Participant

Nelijärve, Estonia

- Fostered open communication and encouragement among a group of 35+ participants.
- Developed business concepts focusing on social entrepreneurship using the *Business Canvas Model*.

AWARDS AND ACHIEVEMENTS

- 2024 **Granted of 9-month program; top 10% of 500+ applicants for technical skills, Fatima Fellowship Dean's Honor Award, Hacettepe University**
- 2023 **Achievement Award, Aselsan**
- 2020 **Best Course Project Award, CENG3204 Artificial Intelligence**
- 2019 **Best Movie Award, Erasmus+ Project (Interstellar Inclusion)**
- 2018 **Awarded \$6,000 for a summer internship grant at Michigan State University, KFAU**
- 2016-21 **Dean's Honor Award, KFAU**
2nd highest rank in the faculty, KFAU
Recipient of merit-based full scholarship and monthly stipend by Dean's Office, KFAU

LANGUAGES

- Fluent in English (written and spoken)
- Beginner in German

ADDITIONAL EXPERIENCE

KFAU Robotics Club, Member of Board

Jan 2019 - Sept 2021

IEEE KFAU Student Branch, Founding President

Jan 2018 - Oct 2019